

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Case Study

COURSE NAME: COMPUTER VISION

COURSE CODE:ITA0515

Slot B

COURSE OUTCOME COVERED

CO3: Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)

Assessment Tool Weightage: 40%

QUESTIONS:5

Total Marks 50

Duration :60 Minutes

Annexure A

Case Study

Code of Conduct:

I PAVAN JANA (192472313) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Pavan j

Rubrics for Evaluation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks(100)
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

1. Preprocessing and Feature Detection

A vision system processes images captured under varying lighting conditions. Without preprocessing, feature detection results are inconsistent. Analyze the issue and explain how preprocessing improves feature detection performance.

Preprocessing and Feature Detection

Analysis

- Images captured under different lighting conditions may have different **brightness and contrast**.
- Shadows, reflections, and poor illumination can hide important image features.
- Noise may also be present in the captured images.
- Because of these variations, the same object may produce **different feature detection results**.

How preprocessing helps

- **Noise removal:** Removes unwanted pixels and improves image quality.
- **Contrast enhancement:** Makes important details more visible.
- **Histogram equalization:** Improves the distribution of brightness levels.
- **Image normalization:** Makes images more consistent under different lighting conditions.
- **Smoothing:** Reduces small unwanted variations before feature detection.

Result

- Features such as **corners, edges, and key points** become clearer.
- The detector can identify features more accurately.
- The same features can be detected more consistently in different images.

Conclusion: Preprocessing reduces noise and lighting variations, improving the accuracy and consistency of feature detection.

2. Edge vs Corner Detection

In an image containing both straight edges and textured regions, a system uses edge detection but fails to identify key points for matching. Analyze why edge detection is insufficient and justify the use of corner detection in this case.

Edge vs Corner Detection

Analysis

- **Edge detection** identifies locations where there is a sudden change in image intensity.
- It is useful for finding object boundaries and straight lines.
- However, an edge usually contains many similar points.
- Therefore, it may be difficult to identify one **unique point** for image matching.
- In textured regions, many edges may also be present, making matching difficult.

Why corner detection is better

- A corner is a point where intensity changes in **more than one direction**.
- Corners are more distinctive than ordinary edge points.
- They can be easily identified again in another image.
- Corner points are useful for:
 - Image matching
 - Object recognition
 - Image stitching

- Tracking
- Motion estimation

Example

- Consider a **window** in an image.
- Edge detection identifies the four sides of the window.
- Corner detection identifies the **four specific points where the sides meet**.
- These points are more useful for matching.

Conclusion: Edge detection is useful for identifying boundaries, but corner detection is better for finding distinctive key points required for image matching.

3. Hough Transform in Noisy Images

A system uses Hough Transform to detect lines in noisy images but produces inaccurate results. Analyze the effect of noise on detection and suggest improvements based on preprocessing or parameter selection.

Hough Transform in Noisy Images

Analysis

- The Hough Transform is commonly used to detect **lines, circles, and other shapes**.
- It works by collecting votes from image points.
- In a noisy image, unwanted pixels can also vote for incorrect lines.
- This creates **false peaks** in the Hough accumulator.
- As a result, incorrect or multiple lines may be detected.

Problems caused by noise

- False line detection

- Incorrect line position
- Extra lines
- Broken or incomplete lines
- Reduced detection accuracy

Improvements

- Apply **Gaussian or median filtering** to reduce noise.
- Use a good **edge detection method** before Hough Transform.
- Adjust the **Hough threshold** to remove weak false detections.
- Set a suitable **minimum line length**.
- Set an appropriate **maximum line gap**.
- Use morphological operations to remove small unwanted regions.

Result

After preprocessing and proper parameter selection:

- Noise is reduced.
- False lines are minimized.
- Important lines receive stronger votes.
- Detection becomes more accurate.

Conclusion: Noise can produce false Hough detections. Noise filtering, better edge detection, and proper Hough parameters improve the accuracy of line detection.

4. Feature Matching under Transformations

Two images of the same object are taken at different scales and orientations. Traditional feature matching fails, but scale invariant feature matching works effectively. Analyze why scale invariant methods perform better in this scenario.

Feature Matching under Transformations

Analysis

- The same object can appear different when the camera position changes.
- The object may become:
 - Larger or smaller
 - Rotated
 - Shifted
 - Viewed from a different orientation
- Traditional feature matching may not recognize the same feature after these changes.

Why scale-invariant methods work better

- Scale-invariant methods detect features that remain recognizable at **different sizes**.
- They create feature descriptions based on local image patterns.
- These descriptions are less affected by scale and rotation.
- Therefore, corresponding points can still be matched between two images.

Example

Suppose an image contains a **car**:

- Image 1: Car appears large.
- Image 2: Car appears small and slightly rotated.
- Traditional matching may fail.

- Scale-invariant methods can still identify common features such as **corners, shapes, and local patterns**.

Advantages

- Better matching accuracy
- Handles different image scales
- Handles rotation
- More robust to viewpoint changes
- Useful for object recognition and image stitching

Conclusion: Scale-invariant feature methods perform better because they can identify corresponding features even when the object's size and orientation change.

5. PCA for Feature Optimization

A system extracts a large number of features, leading to high computational cost and redundancy. Analyze how Principal Component Analysis helps in reducing dimensionality and improving efficiency.

PCA for Feature Optimization

Analysis

- Feature extraction can generate hundreds or thousands of features.
- Some features may contain similar or repeated information.
- Processing all features increases:
 - Computation time
 - Memory usage
 - Processing complexity
- Redundant features can also reduce the efficiency of a machine learning or vision system.

How PCA helps

- **Principal Component Analysis (PCA)** reduces the number of dimensions.
- It identifies the directions containing the **maximum amount of information or variance**.
- The original features are transformed into a smaller number of principal components.
- Less important components can be removed.

Example

Suppose a system extracts **100 features** from an image.

- Many of these features may be correlated.
- PCA may reduce them to **20 important components**.
- The system then works with 20 components instead of 100.

Benefits

- Reduces dimensionality
- Removes redundancy
- Reduces memory usage
- Decreases computation time
- Makes feature processing faster
- Can improve model efficiency

Conclusion: PCA reduces a large feature set into a smaller set of important components while preserving most useful information, making the vision system faster and more efficient.